



EarthSafe

Real-Time Earthquake Hazard Prediction

STAT 418 · UCLA · Spring 2026 · Yutong Ma



earthsafe-eu9o2sytyvmwqnl548ppig7.streamlit.app



Flask API · [POST /predict](#) · [GET /recent](#)

Project Overview & Motivation

The Problem

Every year ~55,000 earthquakes of $M \geq 2.5$ occur globally. USGS publishes the data in real time — but it's raw, technical numbers most people cannot act on.

EarthSafe turns seismic parameters into a clear hazard level: Low, Moderate, or High.

55,000+

$M \geq 2.5$ quakes
per year (global)

2,000

records collected
USGS 2023-2024

5

input features
for prediction

Hazard Levels

 **Low**

Magnitude < 4.0 · Minor shaking, rarely felt

 **Moderate**

Magnitude 4.0–6.0 · Widely felt, possible minor damage

 **High**

Magnitude ≥ 6.0 · Strong shaking, significant damage

Data Collection & Preprocessing

Data Pipeline

01

USGS API Fetch

Monthly chunks · $M \geq 2.5$ · 2023–2024 · 2,000 records

02

Feature Engineering

location_type from place text · impute gap (103 nulls with median)

03

Hazard Labeling

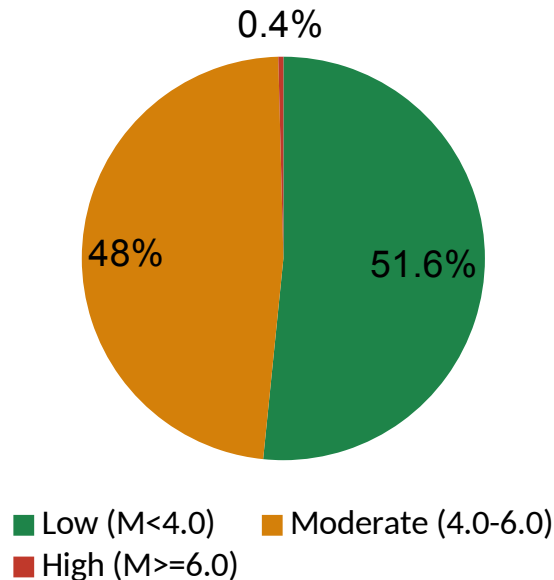
Rule-based: Low < 4.0 · Moderate 4–6 · High ≥ 6.0

04

Stratified CV Split

5-fold cross-validation, stratified by hazard class

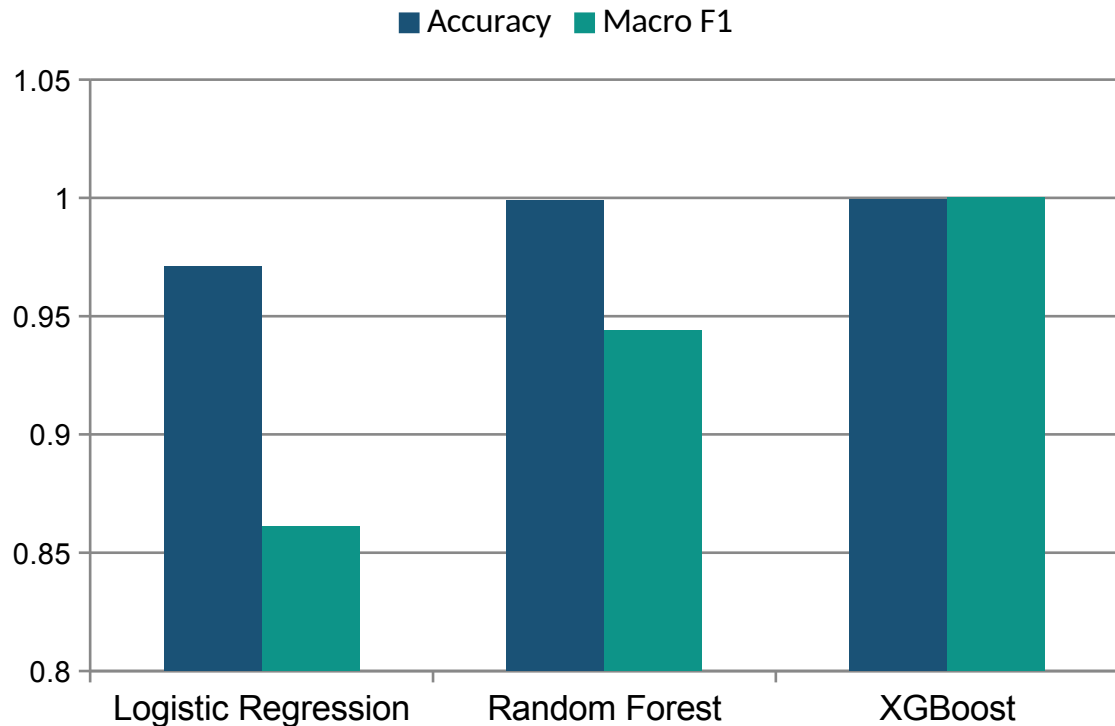
Class Distribution



⚠ High class: only 7 records ($M \geq 6.0$ events are rare).
Handled with balanced sample weights in training.

Model Development & Evaluation

5-Fold Cross-Validation Results



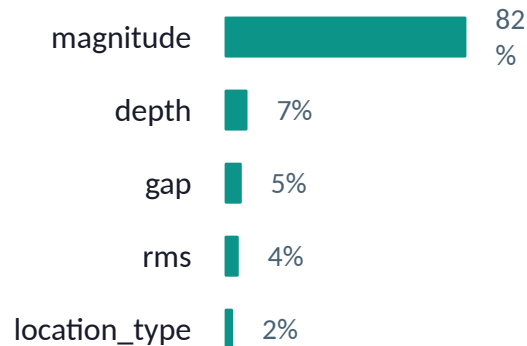
Best Model

XGBoost

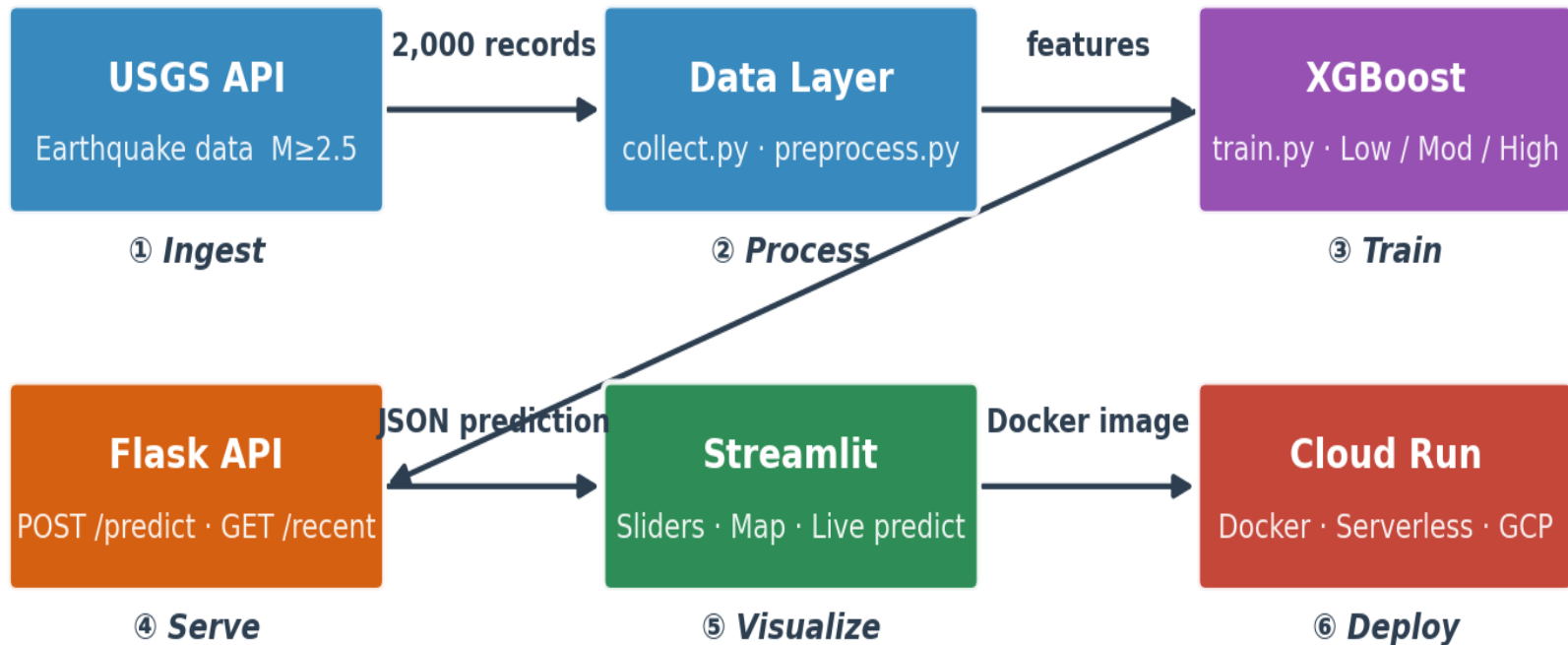
Accuracy 99.95% • Macro F1 1.00

300 trees • depth 6 • lr 0.05 • sample_weight
balanced

Feature Importance



Solution Architecture



Python 3.11

XGBoost 2.0

Flask 3.0

Streamlit 1.35

Plotly 5.20

Docker

Cloud Run

App Features & Live Demo



Interactive Sliders

Drag magnitude, depth, gap, RMS, location type — hazard level updates instantly



Live Earthquake Map

Plotly globe · live USGS data · color-coded by hazard · auto-refreshes every 5 min



Probability Display

Confidence scores for each class + magnitude gauge chart



[Flask API · POST /predict · GET /recent](#)



earthsafe-eu9o2sytymwqnls48ppjg7.streamlit.app



Flask REST API

```
POST /predict
{"magnitude": 7.0, "depth": 10,
 "gap": 90, "rms": 0.3, "location_type":
 0}

→ {"hazard_label": "High",
   "color": "#e74c3c",
   "probabilities": {...}}
```



USGS Live Proxy

GET /recent — last 24h quakes for the map



Docker + Cloud Run

Containerised · serverless · live June 2-9



Auto-Refresh

Map updates every 5 min via cache_data TTL

Challenges, Learnings & Future Work



Challenges

Class Imbalance — 7 High records out of 2,000
Port 5000 → 5001 — macOS AirPlay conflict
brew install libomp — XGBoost on Mac



Learnings

End-to-End MLOps — data → model → API → cloud
Macro F1 > Accuracy — right metric for imbalanced classes
Claude AI — code, debug & deploy at every layer



Future Work

Temporal Features — aftershock sequences
USGS WebSocket — sub-second map updates
Magnitude Regression — early warning potential